Cross-Scanner Robustness for Mitosis Detection

Demo 1 — MIDOG++ human breast cancer subset

Question

Does a model trained on some whole-slide scanners still work on a scanner it has never seen, and does color/stain augmentation help?

Data

MIDOG++ human breast cancer subset: 150 slides from 3 scanners.

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Aperio CS2       50 slides
Hamamatsu S360   50 slides
Hamammatsu XR    50 slides
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Tumor type is held fixed at human breast cancer, so scanner is the only domain shift being tested. Each experiment trains on two scanners and tests on the third (leave-one-scanner-out).

What I Did

Part 1 — Patch classification

Cropped 128×128 patches centered on each annotated candidate and trained classifiers to separate mitotic from non-mitotic candidates:

1. Summary color/intensity features + logistic regression
2. Downsampled 32×32 raw pixels + logistic regression
3. Same, plus generic brightness/contrast/RGB augmentation
4. Same, plus augmentation guided by measured per-scanner RGB means
5. All of the above across all three scanner holdouts

Part 2 — Full detection

Moved from patch classification to actual detection:

6. Built COCO splits using only mitotic boxes as targets
7. Cut 512×512 tiles with balanced positive/negative sampling (2078 train, 1376 test)
8. Trained Faster R-CNN (ResNet50-FPN) on the Aperio CS2 holdout, with and without color jitter
9. Evaluated precision/recall/F1 at IoU 0.3 across a score-threshold sweep

Results

Patch classification (mean across 3 scanner holdouts)

Model	Mean F1	Mean Recall
Pixel baseline	0.429	0.508
Pixel + generic aug	0.463	0.564
Pixel + scanner-stat aug	0.442	0.535

Augmentation helped on two of three holdouts and was flat on Hamammatsu XR. Generic augmentation beat the scanner-statistics version, so the RGB-mean shift was too crude.

> **IMPORTANT CAVEAT**
>
> All of these are below trivial baselines. Always predicting "mitotic" scores F1 0.52–0.60 depending on the fold, and always predicting "not mitotic" scores higher accuracy than any model. Flattened-pixel logistic regression does not transfer across scanners. This is a real negative result, not a win, and the augmentation gain is movement inside that failure regime.

Holdout	Best model F1	All-positive F1	Model acc.	All-negative acc.
Aperio CS2	0.450	**0.598**	0.540	**0.573**
Hamamatsu S360	0.413	**0.522**	0.533	**0.647**
Hamammatsu XR	0.529	**0.555**	0.457	**0.616**

Detection, held-out Aperio CS2 (IoU 0.3, score threshold 0.70)

Model	Precision	Recall	F1
Faster R-CNN	0.602	0.389	0.473
Faster R-CNN + color jitter	**0.604**	**0.727**	**0.660**

Color jitter roughly doubled recall at matched precision. Held-out loss agrees independently: 0.0425 with jitter vs 0.0575 without.

Conclusion

The patch-level experiments show that a weak pixel-level model fails to cross the scanner gap entirely. Once the task was framed as real detection with a proper backbone architecture, color/stain augmentation produced a large recall gain on an unseen scanner, which supports the original hypothesis that cross-scanner robustness is partly a color/stain problem.

Limitations

The detection result is promising but not yet demonstrated:

- **Single run per arm, no random seed.** The two detectors differ in weight initialization and data order as well as augmentation, so the gap is not yet separable from seed variance. Needs 3+ seeds per arm.
- **Trained from random init.** No ImageNet pretrained weights were used, which is why absolute F1 sits well below published MIDOG baselines.
- **Threshold chosen on the test set**, and 0.70 was the highest value swept. The jitter model's F1 was still rising there, so its true optimum was never measured.
- **Recall denominator inflated 1.35×.** Overlapping tiles duplicate 180 of the 688 test mitoses, so reported recall is per-tile-instance rather than per-mitosis.
- **One of three holdouts.** Detection was only run on Aperio CS2, the fold where augmentation looked best at patch level.

MIDOG++ Demo 1 — all figures reproduced from run artifacts under `demo1/models/`.